

SQL Basics Assignment

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Assignment Name: SQL Basics Assignment

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Q1. Create a new database named **company_db** and create a table named **employees** with the following columns:

Column Name	Data Type	Constraint
employee_id	INT	PRIMARY KEY
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
department	VARCHAR(50)	
salary	INT	
hire_date	DATE	

Answer :

```
CREATE DATABASE company_db;
```

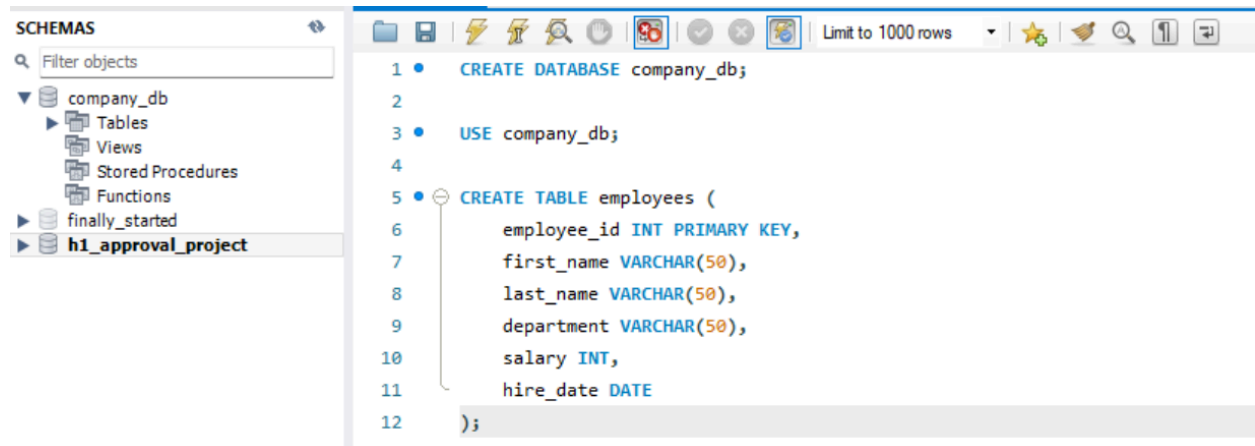
```
USE company_db;
```

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50),
```

```

last_name VARCHAR(50),
department VARCHAR(50),
salary INT,
hire_date DATE
);

```



Q2. Insert the following sample records into the **employees** table.

employee_id	first_name	last_name	department	salary	hire_date
101	Amit	Sharma	HR	50000	2020-01-15
102	Riya	Kapoor	Sales	75000	2019-03-22
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
105	Arjun	Singh	Finance	60000	2022-02-10

Answer :

```

INSERT INTO employees
(employee_id, first_name, last_name, department, salary, hire_date)
VALUES
(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),
(102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),

```

```
(103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),
(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),
(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');
```

```
15 • INSERT INTO employees
16 (employee_id, first_name, last_name, department, salary, hire_date)
17 VALUES
18 (101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),
19 (102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),
20 (103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),
21 (104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),
22 (105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');
23
24 • SELECT * FROM employees;
25
```

Result Grid						
Filter Rows:						
Edit:   						
Export/Import:  						
	employee_id	first_name	last_name	department	salary	hire_date
▶	101	Amit	Sharma	HR	50000	2020-01-15
	102	Riya	Kapoor	Sales	75000	2019-03-22
	103	Raj	Mehta	IT	90000	2018-07-11
	104	Neha	Verma	IT	85000	2021-09-01
	105	Arjun	Singh	Finance	60000	2022-02-10

Q3. Display all employee records sorted by **salary (lowest to highest)**.

Hint: Use the **ORDER BY** clause on the salary column.

Answer :

```
SELECT *
FROM employees
ORDER BY salary ASC;
```

```
26 • SELECT *
27 FROM employees
28 ORDER BY salary ASC;
29
30
```

Result Grid						
Filter Rows:						
Edit:   						
Export/Import: 						
	employee_id	first_name	last_name	department	salary	hire_date
▶	101	Amit	Sharma	HR	50000	2020-01-15
	105	Arjun	Singh	Finance	60000	2022-02-10
	102	Riya	Kapoor	Sales	75000	2019-03-22
	104	Neha	Verma	IT	85000	2021-09-01
	103	Raj	Mehta	IT	90000	2018-07-11

Q4. Show employees sorted by department (A–Z) and salary (High → Low).

Answer :

```
SELECT *  
FROM employees  
ORDER BY department ASC, salary DESC;
```

```
30 • SELECT *  
31 FROM employees  
32 ORDER BY department ASC, salary DESC;  
33
```

Result Grid						
Filter Rows:						
Edit: Export/Import:						
employee_id	first_name	last_name	department	salary	hire_date	
105	Arjun	Singh	Finance	60000	2022-02-10	
101	Amit	Sharma	HR	50000	2020-01-15	
103	Raj	Mehta	IT	90000	2018-07-11	
104	Neha	Verma	IT	85000	2021-09-01	
102	Riya	Kapoor	Sales	75000	2019-03-22	

Q5. List all employees in the IT department, ordered by hire date (Newest First).

Answer :

```
SELECT *  
FROM employees  
WHERE department = 'IT'  
ORDER BY hire_date DESC;
```

```
34 • SELECT *  
35 FROM employees  
36 WHERE department = 'IT'  
37 ORDER BY hire_date DESC;  
38
```

Result Grid						
Filter Rows:						
Edit: Export/Import:						
employee_id	first_name	last_name	department	salary	hire_date	
104	Neha	Verma	IT	85000	2021-09-01	
103	Raj	Mehta	IT	90000	2018-07-11	
* NULL	NULL	NULL	NULL	NULL	NULL	

Q6. Create a table **sales** to track sales data and populate it with the following records.

sale_id	customer_name	amount	sale_date
1	Aditi	1500	2024-08-01
2	Rohan	2200	2024-08-03
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
5	Rohan	4500	2024-09-25

Answer :

```
CREATE TABLE sales (  
    sale_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    amount INT,  
    sale_date DATE  
);
```

```
INSERT INTO sales  
(sale_id, customer_name, amount, sale_date)  
VALUES  
(1, 'Aditi', 1500, '2024-08-01'),  
(2, 'Rohan', 2200, '2024-08-03'),  
(3, 'Aditi', 3500, '2024-09-05'),  
(4, 'Meena', 2700, '2024-09-15'),  
(5, 'Rohan', 4500, '2024-09-25');
```

```
39 CREATE TABLE sales (  
40     sale_id INT PRIMARY KEY,  
41     customer_name VARCHAR(50),  
42     amount INT,  
43     sale_date DATE  
44 );  
45  
46 INSERT INTO sales  
47 (sale_id, customer_name, amount, sale_date)  
48 VALUES  
49 (1, 'Aditi', 1500, '2024-08-01'),  
50 (2, 'Rohan', 2200, '2024-08-03'),  
51 (3, 'Aditi', 3500, '2024-09-05'),  
52 (4, 'Meena', 2700, '2024-09-15'),  
53 (5, 'Rohan', 4500, '2024-09-25');  
54  
55 SELECT * FROM sales;  
56
```

sale_id	customer_name	amount	sale_date
1	Aditi	1500	2024-08-01
2	Rohan	2200	2024-08-03
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
5	Rohan	4500	2024-09-25
*	NULL	NULL	NULL

Q7. Display all sales records sorted by amount (Highest → Lowest).

Hint: Use **ORDER BY amount DESC**.

Answer :

```
SELECT *  
FROM sales  
ORDER BY amount DESC;  
SELECT *  
FROM sales  
ORDER BY amount DESC;
```

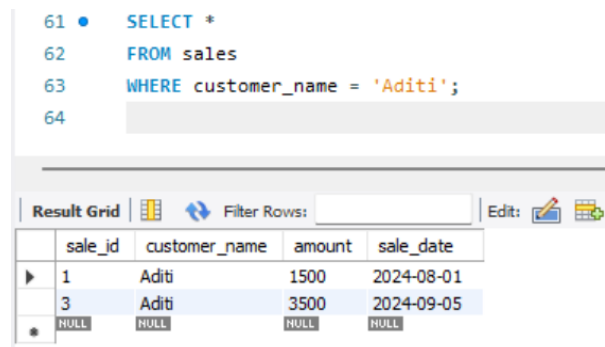
```
57 SELECT *  
58 FROM sales  
59 ORDER BY amount DESC;  
60
```

sale_id	customer_name	amount	sale_date
5	Rohan	4500	2024-09-25
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
2	Rohan	2200	2024-08-03
1	Aditi	1500	2024-08-01
*	NULL	NULL	NULL

Q8. Show all sales made by the customer "Aditi".
Hint: Use `WHERE customer_name = 'Aditi'`.

Answer :

```
SELECT *  
FROM sales  
WHERE customer_name = 'Aditi';
```



```
61 • SELECT *  
62 FROM sales  
63 WHERE customer_name = 'Aditi';  
64
```

	sale_id	customer_name	amount	sale_date
▶	1	Aditi	1500	2024-08-01
	3	Aditi	3500	2024-09-05
•	NULL	NULL	NULL	NULL

Q9. What is the difference between a **Primary Key** and a **Foreign Key**?

Answer :

A Primary Key is a column or a set of columns that uniquely identifies each record in a table. It does not allow duplicate or `NULL` values.

A Foreign Key is a column that creates a relationship between two tables by referring to the Primary Key of another table. It helps maintain referential integrity and ensures that related data remains consistent across tables.

Q10. What are constraints in SQL and why are they used?

Answer :

SQL constraints are rules applied to table columns to ensure the accuracy, consistency, and integrity of data stored in a database.

Constraints help maintain data quality and prevent invalid or inconsistent data from being entered into the database.

Some common SQL constraints are:

- PRIMARY KEY – Uniquely identifies each record.
- FOREIGN KEY – Maintains relationships between tables.
- NOT NULL – Prevents `NULL` values.
- UNIQUE – Ensures all values in a column are different.
- CHECK – Restricts values based on a specified condition.

